

HealthCheck AI - Project Documentation

1. Project Overview

HealthCheck AI is a lightweight, full-stack medical symptom checker. It leverages Machine Learning (Random Forest) to analyze user-provided symptoms and predict potential diseases. The application is designed with a modern glassmorphism UI.

2. Tech Stack

- Backend: FastAPI (Python)
- Machine Learning: Scikit-Learn (Random Forest)
- Frontend: HTML5, CSS3, JavaScript (Vanilla)
- Database: Browser LocalStorage (History Tracking)

3. Key Features

1. Symptom Input: Tag-based selection with suggestions.
2. AI Prediction: Predicts diseases with confidence %.
3. Clinical Metadata: Shows severity, precautions, and health tips.
4. Local History: Stores past diagnostics locally.

4. Installation & Setup

1. Ensure Python 3.8+ is installed.
2. Run: `pip install fastapi uvicorn scikit-learn pandas numpy`
3. Run the project: `./run.ps1`
4. Access at: `http://localhost:3000`

5. File Structure

- backend/: Logic and ML models
- frontend/: Web interface
- data/: CSV and Metadata
- run.ps1: Startup script

6. Disclaimer

For educational purposes only. Not a medical diagnosis tool. Consult a doctor for medical advice.